

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; qwen3.6-35b-a3b

Abstract

This report evaluates `qwen/qwen3.6-35b-a3b` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 0/4 routing modes; OpenRouter leads in 3/4; 1/4 tie; Infron leads Observed cost in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie; Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads E2E latency in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie.

Overall, Infron does not show a clear cross-mode strength in this run, while OpenRouter's cross-mode strengths are throughput, Streaming TTFT, E2E latency, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput OpenRouter wins 3/4 Max advantage 16.86%, higher is better	Observed cost Infron wins 2/4 Max advantage 83.20%, lower is better	E2E latency OpenRouter wins 3/4 Max advantage 3.76%, lower is better	Streaming TTFT OpenRouter wins 3/4 Max advantage 126.67%, lower is better	Cache hit rate OpenRouter wins 3/4 Max advantage 0.00%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 95.05%	Infron advantage 83.20%	OpenRouter advantage 54.47%	OpenRouter advantage 54.06%	OpenRouter advantage 0.00%
Price First price	OpenRouter advantage 7.36%	OpenRouter advantage 6.91%	Infron advantage 17.04%	Infron advantage 10.66%	Tie tie
Latency First latency	OpenRouter advantage 63.65%	Infron advantage 69.65%	OpenRouter advantage 31.46%	OpenRouter advantage 30.13%	OpenRouter advantage 0.00%
TTFT First ttft	Infron advantage 16.86%	OpenRouter advantage 8.36%	OpenRouter advantage 3.76%	OpenRouter advantage 126.67%	OpenRouter advantage 0.00%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 0/4 routing modes; OpenRouter leads in 3/4; 1/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in 2/4 routing modes; OpenRouter leads in 2/4; 0/4 tie	Overall metrics and provider drill-down
Performance	Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads E2E latency in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)
Price First	Minimize request and token cost	Tie	OpenRouter	OpenRouter	Infron	Infron	Mixed result
Latency First	Minimize full–response waiting time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first–token time	OpenRouter	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

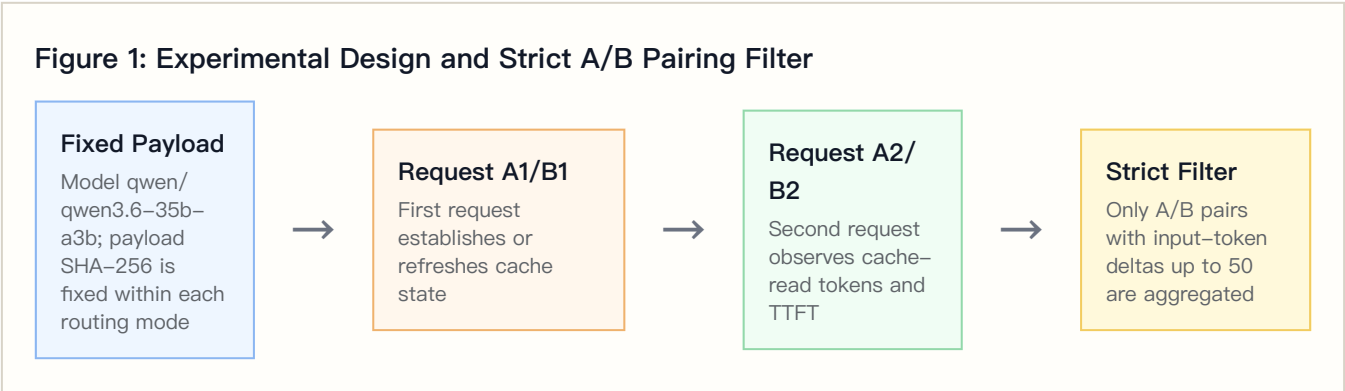
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `business_representative`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	qwen/qwen3.6-35b-a3b
Provider model IDs	infron: qwen/qwen3.6-35b-a3b ; openrouter: qwen/qwen3.6-35b-a3b
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	8
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	0

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	200	7077234	0.00%	\$0.97004100	3.45 tok/s	3669.61 ms	3289.31 ms
Throughput First	OpenRouter	200	7076434	80.73%	\$1.77710850	6.74 tok/s	2375.54 ms	2135.09 ms
Price First	Infron	200	7077234	0.00%	\$1.06600300	4.20 tok/s	3034.01 ms	2685.27 ms
Price First	OpenRouter	200	7076434	0.00%	\$0.99710076	4.51 tok/s	3551.00 ms	2971.46 ms
Latency First	Infron	200	7077234	0.00%	\$1.04751200	4.02 tok/s	3200.40 ms	2835.91 ms
Latency First	OpenRouter	200	7076434	93.45%	\$1.77710850	6.57 tok/s	2434.56 ms	2179.26 ms
TTFT First	Infron	200	7077018	0.00%	\$1.74437200	107.50 tok/s	12666.06 ms	5337.52 ms
TTFT First	OpenRouter	200	7076434	81.66%	\$1.60982660	6.02 tok/s	2658.67 ms	2354.80 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	3216.76 ms	7458.51 ms	10899.25 ms	2873.40 ms	6484.14 ms	8448.48 ms
Throughput First	OpenRouter	2255.99 ms	3644.84 ms	4892.88 ms	2017.72 ms	3474.68 ms	4593.02 ms
Price First	Infron	2889.32 ms	4779.03 ms	6343.18 ms	2572.28 ms	4160.80 ms	5659.26 ms
Price First	OpenRouter	3240.18 ms	5897.34 ms	7736.95 ms	2729.83 ms	5112.79 ms	6511.91 ms
Latency First	Infron	2873.01 ms	5838.77 ms	7245.93 ms	2544.70 ms	5506.94 ms	6732.49 ms
Latency First	OpenRouter	2322.98 ms	3764.15 ms	5126.28 ms	2067.82 ms	3479.26 ms	4806.45 ms
TTFT First	Infron	3454.54 ms	18973.95 ms	355365.91 ms	2905.86 ms	6975.23 ms	125861.03 ms
TTFT First	OpenRouter	2418.94 ms	4494.96 ms	6488.43 ms	2117.27 ms	4264.79 ms	5820.10 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	–2588.14 ms	–2933.10 ms to –2254.04 ms	<0.001	200	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	–2308.45 ms	–2606.72 ms to –2011.13 ms	<0.001	200	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	–3.3780 tok/s	–3.6965 tok/s to –3.0626 tok/s	<0.001	200	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$0.00403534	\$0.00349562 to \$0.00456041	<0.001	200	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	–68.61 pp	–73.28 pp to –63.60 pp	<0.001	200	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	1033.99 ms	803.68 ms to 1275.73 ms	<0.001	200	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	572.37 ms	370.85 ms to 779.48 ms	<0.001	200	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	–0.4959 tok/s	–0.7938 tok/s to –0.2006 tok/s	0.0020	200	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$–0.00034451	\$–0.00039203 to \$–0.00029972	<0.001	200	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	200	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	–1531.69 ms	–1757.15 ms to –1320.45 ms	<0.001	200	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	–1313.30 ms	–1514.04 ms to –1122.51 ms	<0.001	200	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	–2.6790 tok/s	–2.9791 tok/s to –2.3762 tok/s	<0.001	200	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$0.00364798	\$0.00316571 to \$0.00413039	<0.001	200	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	–79.87 pp	–82.87 pp to –76.98 pp	<0.001	200	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	–20014.78 ms	–30552.10 ms to –10915.04 ms	<0.001	200	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	–5965.44 ms	–9738.61 ms to –2849.03 ms	<0.001	200	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	17.0619 tok/s	10.9585 tok/s to 23.8379 tok/s	<0.001	200	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$–0.00067273	\$–0.00287011 to \$0.00132153	0.5616	200	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	–69.17 pp	–73.80 pp to –64.54 pp	<0.001	200	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	0	0.0000	0	0.00 ms	3289.31 ms	3669.61 ms
Throughput First	OpenRouter	6300	15.7500	400	2135.09 ms	2135.09 ms	2375.54 ms
Price First	Infron	0	0.0000	0	0.00 ms	2685.27 ms	3034.01 ms
Price First	OpenRouter	6197	15.4925	400	2971.46 ms	2971.46 ms	3551.00 ms
Latency First	Infron	0	0.0000	0	0.00 ms	2835.91 ms	3200.40 ms
Latency First	OpenRouter	6252	15.6300	400	2179.26 ms	2179.26 ms	2434.56 ms
TTFT First	Infron	539066	1347.6650	85	9941.75 ms	5337.52 ms	12666.06 ms
TTFT First	OpenRouter	6228	15.5700	400	2354.80 ms	2354.80 ms	2658.67 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	100.00%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3-E2:
Normalized
capability
contour**

Five metrics are normalized to 0-100, where higher is better.

**Figure 3-E3:
Cost-cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure 3-E1:
Routing-mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

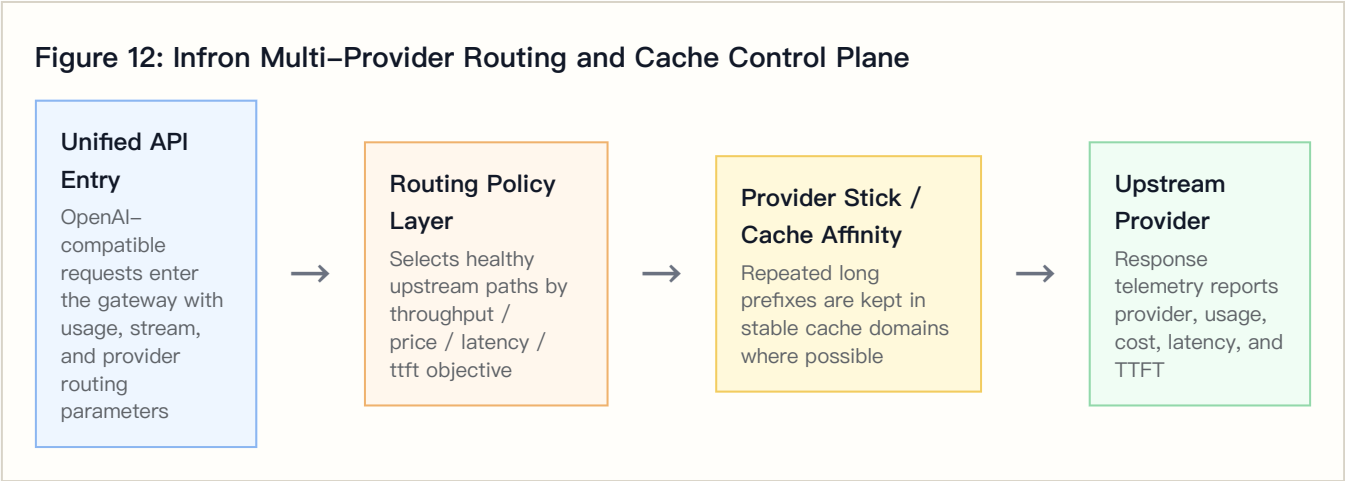
Figure 4–E3: Latency First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	400	400	alibaba/eu 200, deepinfra 193, alibaba/us 7

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	400	400	WandB 400
Price First	Infron	400	400	deepinfra 396, alibaba/eu 4
Price First	OpenRouter	400	400	AkashML 400
Latency First	Infron	400	400	deepinfra 354, alibaba/eu 46
Latency First	OpenRouter	400	400	WandB 400
TTFT First	Infron	400	400	deepinfra 292, atlas-cloud 85, siliconflow 23
TTFT First	OpenRouter	400	400	WandB 320, Parasail 80

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	alibaba/eu	200	50.00%	101/99	112	3465.66 ms	3717.07 ms	3507843	2537
Throughput First	Infron	deepinfra	193	48.25%	97/96	110	3033.64 ms	3554.49 ms	3390643	2446
Throughput First	Infron	alibaba/us	7	1.75%	2/5	7	5299.82 ms	5487.71 ms	178748	86
Throughput First	OpenRouter	WandB	400	100.00%	200/200	200	2135.09 ms	2375.54 ms	7076434	6400
Price First	Infron	deepinfra	396	99.00%	197/199	199	2689.20 ms	3038.84 ms	7060933	5042
Price First	Infron	alibaba/eu	4	1.00%	3/1	3	2296.83 ms	2556.01 ms	16301	51
Price First	OpenRouter	AkashML	400	100.00%	200/200	200	2971.46 ms	3551.00 ms	7076434	6400
Latency First	Infron	deepinfra	354	88.50%	175/179	179	2746.71 ms	3124.66 ms	6352042	4548
Latency First	Infron	alibaba/eu	46	11.50%	25/21	25	3522.38 ms	3783.27 ms	725192	593
Latency First	OpenRouter	WandB	400	100.00%	200/200	200	2179.26 ms	2434.56 ms	7076434	6400
TTFT First	Infron	deepinfra	292	73.00%	146/146	150	3645.28 ms	3978.48 ms	5123026	3728
TTFT First	Infron	atlas-cloud	85	21.25%	43/42	47	4196.88 ms	37211.25 ms	1520260	540565

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
TTFT First	Infron	siliconflow	23	5.75%	11/12	13	31037.04 ms	32250.11 ms	433732	368
TTFT First	OpenRouter	WandB	320	80.00%	163/157	168	2304.99 ms	2568.13 ms	5982335	5120
TTFT First	OpenRouter	Parasail	80	20.00%	37/43	48	2554.05 ms	3020.86 ms	1094099	1280

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache–hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	–80.73 pp	0.55x	alibaba/eu 50.00%	WandB 100.00%	–6300	Cache and cost move in different directions; evaluate with speed metrics
Price First	+0.00 pp	1.07x	deepinfra 99.00%	AkashML 100.00%	–6197	Infron has higher cache but higher cost; inspect upstream unit price and completion/ reasoning tokens
Latency First	–93.45 pp	0.59x	deepinfra 88.50%	WandB 100.00%	–6252	Cache and cost move in different directions; evaluate with speed metrics
TTFT First	–81.66 pp	1.08x	deepinfra 73.00%	WandB 80.00%	+532838	OpenRouter has higher cache and lower cost; provider/cache–domain mix is the main signal

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second–request cache–read tokens, token–level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt–length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	268	534144	0	0.00%	\$0.19801600	3368.60 ms	2550.69 ms

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	OpenRouter	268	533644	189024	35.42%	\$0.23677628	1923.01 ms	1589.94 ms
medium	8000	Infron	268	2767942	0	0.00%	\$1.18726400	6726.97 ms	3294.91 ms
medium	8000	OpenRouter	268	2767440	1720224	62.16%	\$1.20359530	2651.55 ms	2304.47 ms
long	32000	Infron	264	10852274	0	0.00%	\$3.44264800	6850.02 ms	4784.02 ms
long	32000	OpenRouter	264	10851784	7142784	65.82%	\$4.72077278	3704.45 ms	3350.07 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	0.00%	43.53%	OpenRouter
short	Price First	0.00%	0.00%	tie
short	Latency First	0.00%	53.03%	OpenRouter
short	TTFT First	0.00%	45.12%	OpenRouter
medium	Throughput First	0.00%	79.67%	OpenRouter
medium	Price First	0.00%	0.00%	tie
medium	Latency First	0.00%	90.66%	OpenRouter
medium	TTFT First	0.00%	78.30%	OpenRouter
long	Throughput First	0.00%	82.83%	OpenRouter
long	Price First	0.00%	0.00%	tie
long	Latency First	0.00%	96.14%	OpenRouter
long	TTFT First	0.00%	84.31%	OpenRouter

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.22359400	7702.35 ms	7308.90 ms
Infron	2	50	50	0.00%	\$0.25005600	9135.10 ms	7815.39 ms
Infron	3	50	50	0.00%	\$0.25737700	6604.45 ms	5715.52 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	4	50	50	0.00%	\$0.23901400	6224.35 ms	5975.82 ms
OpenRouter	1	50	50	60.28%	\$0.43554050	3615.14 ms	3511.92 ms
OpenRouter	2	50	50	79.47%	\$0.45509750	3792.98 ms	3661.66 ms
OpenRouter	3	50	50	89.27%	\$0.45093000	3596.89 ms	3325.91 ms
OpenRouter	4	50	50	93.65%	\$0.43554050	3407.54 ms	3099.22 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.26093300	5236.81 ms	4181.45 ms
Infron	2	50	50	0.00%	\$0.27310200	4460.94 ms	3968.60 ms
Infron	3	50	50	0.00%	\$0.27060000	4617.50 ms	4159.84 ms
Infron	4	50	50	0.00%	\$0.26136800	4561.30 ms	4209.69 ms
OpenRouter	1	50	50	0.00%	\$0.24438268	6053.91 ms	5323.49 ms
OpenRouter	2	50	50	0.00%	\$0.25533460	5388.63 ms	4469.77 ms
OpenRouter	3	50	50	0.00%	\$0.25300080	6123.38 ms	5404.99 ms
OpenRouter	4	50	50	0.00%	\$0.24438268	6286.82 ms	4705.18 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.24239800	6371.33 ms	5948.23 ms
Infron	2	50	50	0.00%	\$0.27311600	4398.27 ms	3965.53 ms
Infron	3	50	50	0.00%	\$0.27061600	6807.02 ms	5670.92 ms
Infron	4	50	50	0.00%	\$0.26138200	4489.10 ms	4124.05 ms
OpenRouter	1	50	50	89.03%	\$0.43554050	3566.86 ms	3405.12 ms
OpenRouter	2	50	50	94.97%	\$0.45509750	3886.87 ms	3738.04 ms
OpenRouter	3	50	50	94.91%	\$0.45093000	3696.95 ms	3300.81 ms
OpenRouter	4	50	50	94.75%	\$0.43554050	3743.07 ms	3482.91 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.26138200	4393.42 ms	4004.04 ms
Infron	2	50	50	0.00%	\$0.27311600	5196.78 ms	4536.92 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	3	50	50	0.00%	\$0.62394200	22840.16 ms	7045.89 ms
Infron	4	50	50	0.00%	\$0.58593200	130005.67 ms	101799.48 ms
OpenRouter	1	50	50	65.52%	\$0.42390620	4494.96 ms	4334.69 ms
OpenRouter	2	50	50	89.38%	\$0.45066250	3702.16 ms	3582.53 ms
OpenRouter	3	50	50	90.44%	\$0.44434730	3742.43 ms	3435.72 ms
OpenRouter	4	50	50	80.62%	\$0.29091060	5246.01 ms	4623.53 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (4/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	Mixed result	High-frequency templates, support automation, marketing, RAG prefixes	Good for throughput-first tasks; constrain cost and cache separately
Latency First	Minimize full-response waiting time	OpenRouter leads overall (4/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	First-token and E2E response are faster; check cache and cost
TTFT First	Minimize streaming first-token time	OpenRouter leads overall (4/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	First-token and E2E response are faster; check cache and cost

11. Conclusion

Routing-Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Price First	Minimize request and token cost	Tie	OpenRouter	OpenRouter	Infron	Infron	Mixed result
Latency First	Minimize full-response waiting time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first-token time	OpenRouter	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short-window stability but not day-level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built-in templates do not cover every workload distribution	Use sanitized production-stratified corpora	Discuss representative long-context templates only
Cost-field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json
Paired dataset	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv
Request-level dataset	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl
Filtered structured records	export/qwen36_35b_a3b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json

Artifact	Path
Excluded-record audit	<code>export/qwen36_35b_a3b_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_records_excluded.json</code>
Test source	<code>tests/test_rerun_routing_sort_cache_cost_ab.py</code>
Benchmark runner source	<code>scripts/rerun_routing_sort_cache_cost_ab.py</code>
HTML report renderer source	<code>scripts/render_glm52_deepseek_style_report.py</code>
Dataset reference	<code>business_representative</code> built-in representative business templates; request-level export is <code>benchmark_requests</code>